

orchestration, it can be the core or access. Even in access, there are more thoughts on how RAN automation is going to happen and how to support the real-time and non-real-time scenarios? In case of services offered, not only the telco-specific service but other digital players also would participate. Everything is getting bundled, whether it is from hyperscalers or vertical solutions.

“Taking the various flavours of orchestrations and now adding an edge into it—for example, data centres getting converted to edge facilities—we are moving the enterprise workloads to the edge, or the 4G offloading local breakout and then content caching. So, the idea on how the orchestration is happening from the edge layer and when moving towards the CPE side, it can be seen as the HOME edge scenario. There is a connection till the end-point, which is the gateway, but if you want to go further and connect with multiple devices at home, how that can be done should be thought about,” explains Gnanapriya.

Based on the kind of requirements, different kinds of flavours can be easily addressed in an automated dynamic orchestration platform fashion, giving enormous facilities and capabilities where these solutions can be leveraged. Specifically in the Indian context, not just the operational efficiencies but also from a cost perspective, bringing all these concepts together actually plays a vital role.

Smart cities stay smart

The smart city concept is gaining momentum with all products coming under a digital shadow. It is a functional and structural improvement of existing cities that strategically uses several smart factors, capitalising on information and communication technology to increase the city's sustainable growth and strengthen functions, while ensuring citizens' enhanced quality of life and health.

Inder points out to one of the open source platforms—India urban

data exchange, which is being deployed across India for sharing data. If the kind of solutions that smart cities are deploying are looked at, there is a layer of software, a middleware layer, which for the most part is non-differentiated. It allows you to find data and provides access to controls, he explains.

“Companies collaboratively work together to create a common open source platform. It has a significant role in the smart cities space. Everyone understands and can deploy and support, but one has to focus on creating value on top,” says Inder.

Bridge gaps with organisational construct

India has announced too many smart cities to be built in different metrics. It has taken a pretty ambitious initiative of building a hundred different smart cities, and this project has been going on for about six years. However, it has not moved quickly, and the results have not been as dramatic as was expected initially.

“It's because a lot of data—records, videos etc—are being collected in these smart cities, but they sit right at the silos or closed proprietary platforms without any well architected interfaces,” says Inder.

It is very difficult for someone who has an innovative idea to get access to this data, or even understand what data is available. For example, say you want to build some kind of emergency response system within a city and would like to take data from the police, fire departments, and hospital services. But you would be unable to do so since it is not possible in these cities. These departments work on completely different systems that have no correlation. Common application programming interfaces (APIs), essentially a middleware enabler, help bridging the gap and creating some kind of platform connecting the systems together and bringing data available into a common format and

making it available.

Inder explains, “This is a prime example where open source can play a role. It is built on national infrastructure, and I believe that anything that is being created for the public using public funds should be open source. It should be done as a collaborative project where multiple parties can work together. We have tried to create a model that is not just open source but also the organisational construct. So, different parties can contribute code and create the governance model for an open source project. This takes a lot of effort and, very often, it is really difficult to maintain and manage because there are always going to be conflicts.”

We have disaggregated the deployment of smart city systems. Previously, a city would go to a vendor who would build the vertically integrated system. But we have realised that it is not the way to build the systems, and that we have to split the systems into horizontal layers. At the bottom, we have the sensors from different vendors. You can mix and match the different sensors as cities now have the ability to be vendor-agnostic. They can make a purchase of sensors, let's say from Samsung, and then buy something from Philips and have the companies compete against each other. The idea here is to be able to do innovation below at the sensor level, data collection level, and then at the application level. “Create applications and services on top. Collaborate in the middle to maximise the availability, and involve more players,” Inder adds.

On a global perspective, the smart city construct is similar all across, and part of the focus is on how the cross-industry solutions are brought together and their automation. For example, how we can support factories of the future and bring in these kinds of technologies together, helping in the automation part, the right kind of connectivity, and bringing the right kind of insights so that one